Exploratory Statistical Analysis of Amazon Data Science Books

Florentia Pattidou Valeria Shirshina Marios Makrygiannis

University of Cyprus
DSC531 – Statistical Simulation and Analysis

Abstract

This report presents a statistical analysis of the Amazon Data Science Books dataset using the R programming language. The objectives are to assess data quality, describe the main characteristics of the books, and investigate how book attributes relate to price and customer ratings.

The original dataset contains 946 observations and 11 variables. During data cleaning, one exact duplicate was removed, blank strings were converted to missing values, and book weights were converted into kilograms. Potential outliers were investigated using the interquartile range (IQR) method, and ten implausible weight measurements were identified using a weight-per-page check and excluded from the weight-specific analysis. A rating consistency check flagged 90 of the 822 reviewed books (10.95%) whose reported average rating differs from the rating implied by the star proportions; these books were retained and documented as a data quality limitation.

Descriptive statistics and visualizations show that prices, page counts and review counts are right-skewed, while average ratings are concentrated near the upper end of the five-star scale. Five hypotheses were tested using Pearson and Spearman correlation tests, Welch's t-test and the Wilcoxon rank-sum test. Page count is positively correlated with log price ($r = 0.486$), books without reviews have a higher mean log price than books with reviews, and review volume is weakly negatively correlated with price ($\rho = -0.156$). No significant association was found between average rating and price, and the difference in ratings between high- and low-engagement books was not robust.

A multiple linear regression for log price explains about 29% of its variation, with page count as the strongest positive predictor and review volume as a negative predictor. Diagnostic plots indicate non-normal residuals and heteroskedasticity, so the regression results should be interpreted with caution.

Contents

1 Introduction **5**

2 Dataset Description **6**

2.1 Dataset Overview 6

2.2 Description of Variables 6

2.3 Initial Data Quality Assessment 6

3 Data Cleaning **7**

3.1 Data Cleaning Overview 7

3.2 Missing Values and Duplicate Observations 7

3.3 Weight Standardization 7

3.4 Outlier Detection Using the IQR Method 7

3.5 Investigation of Suspicious Weight Measurements 8

3.6 Review Data Consistency 9

3.7 Final Cleaned Dataset 10

4 Descriptive Statistics **11**

4.1 Summary of Numerical Variables 11

4.2 Central Tendency and Variability 11

4.3 Distribution Shape and Skewness 11

5 Data Visualization **12**

5.1 Histograms of Numerical Variables 12

5.2 Boxplots and Outlier Visualization 13

5.3 Log Transformation of Review Counts 14

5.4 Log-Transformed Variables and Average Customer Ratings 15

5.5 Star Rating Distribution 17

5.6 Scatterplots and Relationships 20

5.7 Customer Reviews and Ratings 21

5.8 Correlation Matrix Heatmap 22

5.9 Pairwise Scatter Plot Matrix 24

5.10 Categorical Analysis 25

6 Hypothesis Testing and Empirical Findings **26**

6.1 Hypothesis 1: The Effect of Page Count on Book Price and Non-Parametric Modeling 26
6.2 Hypothesis 2: Price Differentiation Based on Review Engagement 28
6.3 Hypothesis 3: Variance in Ratings Across Review Engagement Levels 29
6.4 Hypothesis 4: Review Volume vs. Book Pricing Dynamics 30
6.5 Hypothesis 5: Average Customer Rating vs. Book Price 31

7 Multivariate Econometric Modeling and Diagnostic Evaluation 32
7.1 Multiple Linear Regression Model 32
7.2 Empirical Regression Results 32
7.3 Interpretation of Regression Coefficients 32
7.4 Multicollinearity Assessment 33
7.5 Regression Diagnostic Evaluation 33

8 Conclusion 35
8.1 Summary of Findings 35
8.2 Limitations and Possible Extensions 35

1 Introduction

Data analysis plays an important role in understanding patterns, identifying relationships, and extracting meaningful insights from real-world datasets. Statistical methods allow researchers to summarize large amounts of information, evaluate variability, and investigate potential associations between variables.

This report presents an exploratory statistical analysis of the Amazon Data Science Books dataset using the R programming language. The dataset contains information about books related to data science, including their prices, number of pages, average customer ratings, number of reviews, rating distributions, and physical weights.

The study has three objectives. First, to assess and improve data quality by identifying missing values, duplicate observations, inconsistent units, potential outliers and inconsistencies between rating variables. Second, to describe the distributions of the main variables using descriptive statistics and graphical methods. Third, to investigate relationships between book characteristics, prices and customer ratings using formal hypothesis tests and a multiple linear regression model.

In particular, the analysis addresses the following questions:

- Are longer books more expensive?
- Do books with and without customer reviews differ in price?
- Do books with many reviews receive different ratings from books with few reviews?
- Is the number of reviews associated with price?
- Do higher-rated books command higher prices?

The report is organized as follows. Section 2 describes the dataset and Section 3 the data cleaning process. Section 4 presents descriptive statistics and Section 5 the graphical analysis. Section 6 reports the hypothesis tests, Section 7 the regression model and its diagnostics, and Section 8 summarizes the main conclusions and limitations.

2 Dataset Description

2.1 Dataset Overview

The Amazon Data Science Books dataset contains 946 observations and 11 variables describing book characteristics and customer feedback. These include prices, page counts, customer ratings, review counts, star-rating proportions, weights, and languages.

The dataset was imported into R using `read.csv()` and initially examined using `dim()`, `str()`, and `summary()`. This examination identified missing values and inconsistent weight units, which were addressed during data cleaning.

2.2 Description of Variables

Table 1 presents the variables included in the original dataset, their data types, and their meanings.

Variable	Data Type	Description
`price`	Numeric	Book price
`pages`	Integer	Number of pages
`avg_reviews`	Numeric	Average customer rating (1–5)
`n_reviews`	Integer	Number of customer reviews
`star5`	Numeric	Proportion of five-star ratings
`star4`	Numeric	Proportion of four-star ratings
`star3`	Numeric	Proportion of three-star ratings
`star2`	Numeric	Proportion of two-star ratings
`star1`	Numeric	Proportion of one-star ratings
`weight`	Character	Book weight in ounces or pounds
`language`	Character	Language of the book

Table 1: Description of the Original Dataset Variables

The five star-rating variables represent proportions rather than absolute review counts. The weight variable requires conversion to a common measurement unit before numerical analysis.

2.3 Initial Data Quality Assessment

Table 2 summarizes the missing values identified during the initial inspection of the dataset.

Variable	Missing Values
`price`	2
`pages`	12
`avg_reviews`	123
`weight`	3 blank strings
`language`	3 blank strings

Table 2: Initial Missing Values

The initial assessment identified missing information in several variables. Blank strings in the weight and language fields required conversion to `NA`. Missing observations were not automatically removed, allowing available information to be retained for subsequent analysis. Duplicate records and potentially unusual measurements were investigated during the cleaning process described in Section 3.

3 Data Cleaning

3.1 Data Cleaning Overview

Data cleaning was performed in R to improve data consistency while preserving as much information as possible. The procedures included handling missing values, removing duplicate observations, standardizing weight measurements, investigating outliers, and validating customer review information.

3.2 Missing Values and Duplicate Observations

Blank strings in the weight and language variables were converted to `NA`. Missing values were handled using available-case analysis rather than automatically removing incomplete observations, even when multiple values were missing from the same row.

This approach allowed observations to contribute to analyses involving their available variables, minimizing unnecessary information loss. However, available-case analysis may introduce bias if missingness is systematic.

One exact duplicate was identified and removed, reducing the dataset from 946 to 945 observations.

3.3 Weight Standardization

Book weights were originally recorded in two units: ounces (204 observations) and pounds (738 observations), with 3 missing measurements. The following conversion factors were applied:

$$\text{Ounces to kilograms:} \quad W_{\text{kg}} = W_{\text{oz}} \times 0.0283495$$

$$\text{Pounds to kilograms:} \quad W_{\text{kg}} = W_{\text{lb}} \times 0.4535924$$

3.4 Outlier Detection Using the IQR Method

Potential outliers were identified using the interquartile range (IQR) method, with boundaries defined as:

$$IQR = Q_3 - Q_1$$

$$\text{Lower Bound} = Q_1 - 1.5 \times IQR$$

$$\text{Upper Bound} = Q_3 + 1.5 \times IQR$$

For book weight, the IQR was approximately 0.448 kg, producing an upper boundary of 1.602 kg. The method identified 41 potential weight outliers. Other variables also contained outliers: price (97), pages (17), average ratings (34), and review counts (111).

These observations were not automatically removed because extreme values may represent genuine book characteristics rather than recording errors.

3.5 Investigation of Suspicious Weight Measurements

The IQR method identified potential weight outliers, but these observations were not necessarily incorrect. To investigate unusual weight measurements further, the relationship between book weight and page count was examined.

A weight-per-page ratio was calculated for books with available positive page counts and weight measurements:

$$\text{Weight per page (g/page)} = \frac{1000 \times W_{\text{kg}}}{\text{Pages}}$$

Measurements outside the exploratory range of 0.3–10 grams per page were flagged as potentially suspicious. This criterion identified 10 observations.

Criterion	Value
Lower threshold	0.3 g/page
Upper threshold	10 g/page
Flagged observations	10

Table 3: Criteria for Flagging Suspicious Weight Measurements

The flagged measurements were not confirmed data errors. Therefore, the corresponding books were retained in the cleaned dataset. For weight-specific sensitivity analysis, a separate variable, `weight_analysis`, was created by replacing the 10 flagged weight values with `NA`.

The mean weight decreased from approximately 0.778 kg to 0.753 kg, while the standard deviation decreased from 0.583 kg to 0.404 kg. The Pearson correlation between page count and weight increased from 0.571 to 0.865. Because the filtering criterion was based on weight relative to page count, this increase should be interpreted cautiously rather than as independent evidence of a stronger underlying relationship. The corresponding scatterplots are presented in Figure 1.

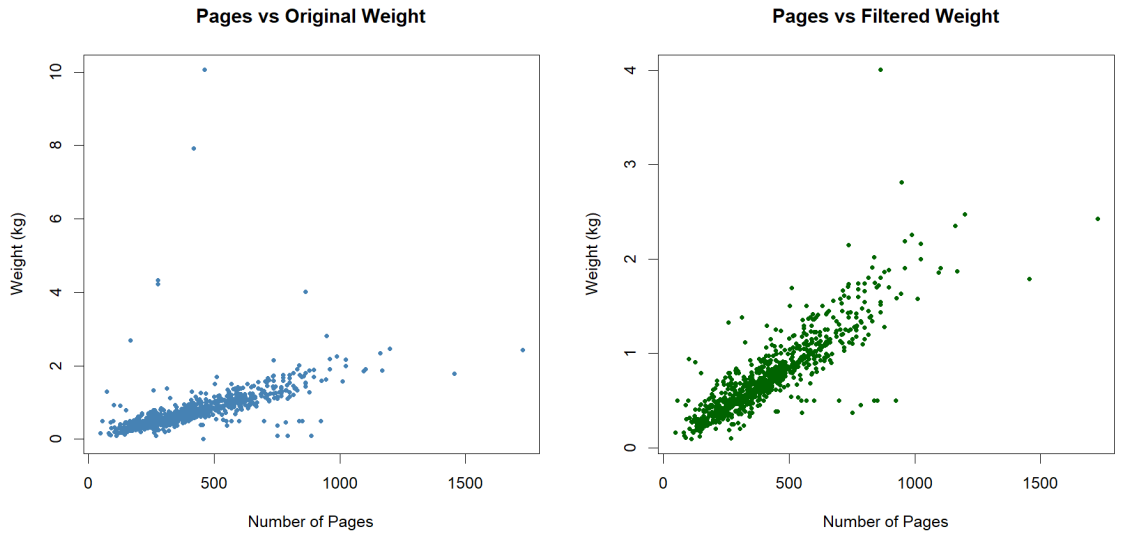


Figure 1: Relationship Between Page Count and Book Weight Before and After Filtering Suspicious Measurements

Figure 1 compares the relationship between page count and book weight before and after excluding the ten flagged measurements from the weight-specific sensitivity analysis.

The original standardized weight measurements include all available values, whereas the filtered analysis excludes potentially suspicious measurements without removing the corresponding books from the dataset.

Although the filtered data show a stronger positive correlation, this increase should be interpreted cautiously because the filtering criterion itself depends on the weight-to-page ratio.

3.6 Review Data Consistency

Customer review information was examined to identify inconsistencies between review counts, average ratings, and star-rating proportions.

A total of 123 books had zero customer reviews. For these observations, the star-rating proportions were set to `NA`, since no rating distribution could be meaningfully calculated.

For the remaining 822 books with customer reviews, the five star-rating proportions were checked to determine whether their sum was approximately equal to one. No observations exceeded a tolerance of 0.05 from the expected total.

An additional consistency check compared the reported average rating with the weighted average calculated from the star-rating proportions:

$$R_{\text{calculated}} = 5s_5 + 4s_4 + 3s_3 + 2s_2 + s_1$$

where s_k represents the proportion of k-star ratings. The absolute difference was calculated as:

$$D = |R_{\text{reported}} - R_{\text{calculated}}|$$

Using an exploratory threshold of $D > 0.2$, 90 out of 822 books (10.95%) were flagged for potential rating inconsistencies.

These discrepancies were concentrated among books with relatively few customer reviews. Some differences exceeded what would ordinarily be expected from rounding. However, the available data do not confirm whether the reported average ratings and star-rating proportions refer to identical sets of customer evaluations.

Since the available data did not establish which rating measure was more reliable, the flagged observations were retained without modification.

Although the star-rating proportions generally summed to one, this additional check revealed potential inconsistencies between the reported average ratings and rating distributions. These should be considered when interpreting rating-related statistical analyses.

3.7 Final Cleaned Dataset

After data cleaning, the dataset contained 945 observations, following the removal of one exact duplicate. Missing values were preserved as `NA`, and available weight measurements were standardized to kilograms.

The cleaned dataset included the original numerical book characteristics and star-rating proportions, together with two weight-related variables: `weight_kg`, containing the standardized measurements, and `weight_analysis`, used for sensitivity analysis after flagging 10 potentially suspicious weight values.

Characteristic	Final Result
Original observations	946
Final observations	945
Duplicate observations removed	1
Available weight measurements	942
Weight values used in sensitivity analysis	932
Flagged suspicious weights	10
Potential rating inconsistencies	90

Table 4: Summary of the Final Cleaned Dataset

The cleaned dataset was exported as a CSV file for subsequent descriptive statistics and visualization. No books were removed solely because of extreme values or missing information.

4 Descriptive Statistics

4.1 Summary of Numerical Variables

Descriptive statistics were calculated for the main numerical variables, including book prices, page counts, average customer ratings, review counts, and filtered book weights.

Table 5 presents the number of available observations, mean, median, standard deviation, coefficient of variation (CV), and skewness.

Variable	N	Mean	Median	SD	CV (%)	Skewness
Price	943	52.111	43.970	36.609	70.25	2.421
Pages	933	419.478	384.000	204.060	48.65	1.123
Average Rating	822	4.441	4.500	0.401	9.03	-2.011
Number of Reviews	945	137.898	25.000	499.731	362.39	9.467
Filtered Weight (kg)	932	0.753	0.669	0.404	53.66	1.670

Table 5: Descriptive Statistics of the Main Numerical Variables

The number of available observations differs between variables because missing values were excluded only from the relevant calculations. Filtered weight statistics were calculated using the `weight_analysis` variable.

4.2 Central Tendency and Variability

The number of customer reviews exhibits the greatest relative variability (CV = 362.39%). Its mean (137.898) is considerably higher than its median (25), indicating that a small number of highly reviewed books influence the average.

Book prices also exhibit substantial variability (CV = 70.25%), while page counts and filtered weights show moderate-to-high relative variability. In contrast, average customer ratings have the lowest CV (9.03%), indicating that ratings are concentrated within a relatively narrow range.

4.3 Distribution Shape and Skewness

Skewness measures the asymmetry of a distribution. Positive values indicate a longer right tail, while negative values indicate a longer left tail.

Customer review counts exhibit strong positive skewness (9.467), indicating that most books receive relatively few reviews, while a small number receive exceptionally high counts. Book prices (2.421), page counts (1.123), and filtered weights (1.670) are also positively skewed.

In contrast, average customer ratings are negatively skewed (-2.011), reflecting the concentration of ratings toward the upper end of the five-star scale.

Original book weights had a skewness of 7.209, which decreased to 1.670 after excluding ten potentially suspicious measurements from the weight-specific analysis.

Overall, the distributions are asymmetric, highlighting the importance of using the median and interquartile range alongside the mean and standard deviation.

5 Data Visualization

5.1 Histograms of Numerical Variables

Histograms were used to examine the distributions of book prices, page counts, customer review counts, average ratings, and filtered book weights.

Figure 2 presents the distributions of these five numerical variables.

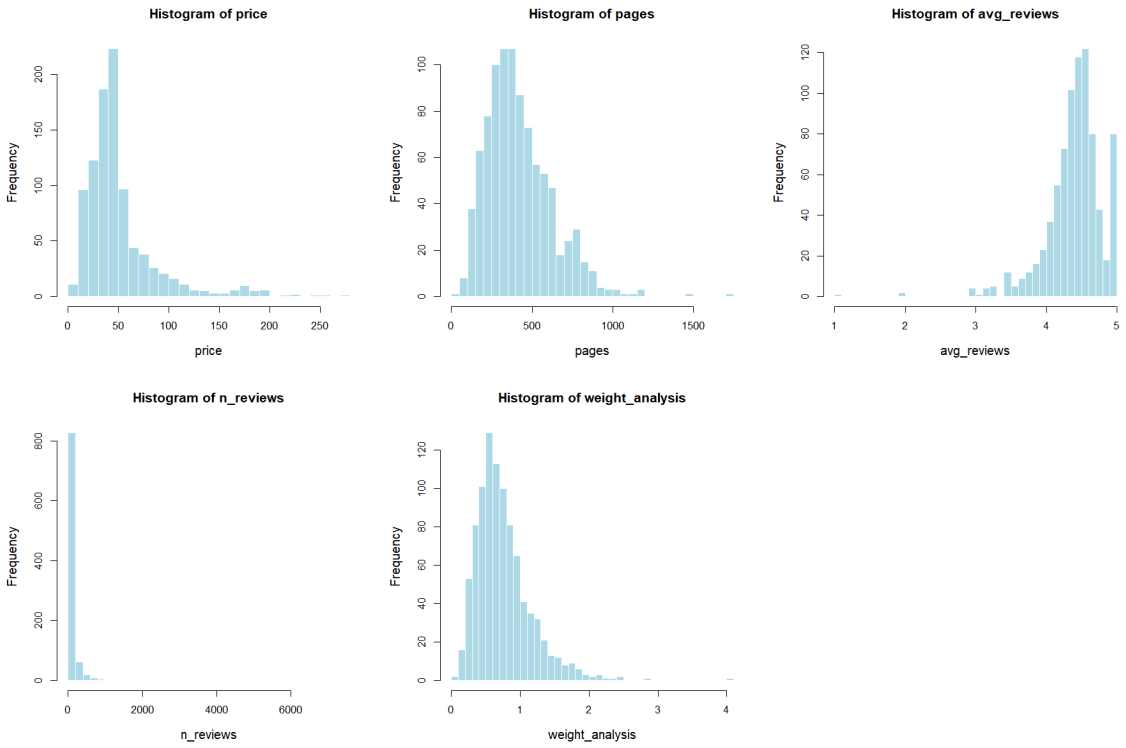


Figure 2: Histograms of the Main Numerical Variables

The histograms show that book prices, page counts, and review counts are positively skewed. Review counts exhibit particularly strong right-skewness, with most books receiving relatively few reviews and a small number receiving thousands.

Average customer ratings are concentrated toward the upper end of the five-star scale, while filtered book weights remain positively skewed. These findings are consistent with the descriptive statistics presented in Section 4.

More specifically, the distribution of average customer ratings is negatively skewed, with most observations concentrated around 4.5 stars.

An additional concentration is visible at the maximum rating of 5.0, indicating that some books have received a perfect average customer rating.

This pattern may be influenced by books with relatively few customer reviews, although the histogram alone cannot establish the underlying explanation.

The concentration of ratings near the upper boundary also highlights the bounded nature of the five-star rating scale.

5.2 Boxplots and Outlier Visualization

Boxplots were used to compare the variability of the main numerical variables and identify observations lying beyond the typical range of each distribution.

Figure 3 presents the boxplots of book prices, page counts, average ratings, review counts, and filtered book weights.

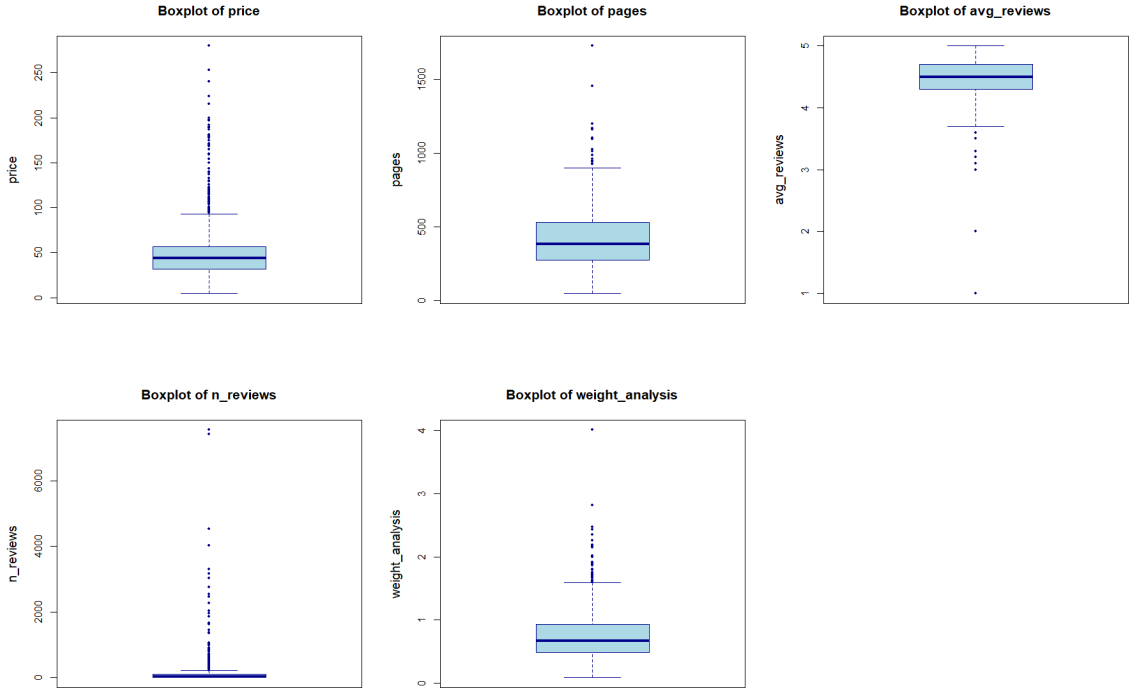


Figure 3: Boxplots of the Main Numerical Variables

The boxplots reveal substantial differences in the spread of the numerical variables. Review counts exhibit numerous extreme observations, consistent with their high standard deviation and coefficient of variation.

Book prices and page counts also contain observations beyond the upper whiskers, indicating the presence of relatively expensive and unusually long books.

Average customer ratings are concentrated near the upper end of the rating scale, with some comparatively low ratings appearing as outliers. Filtered book weights still contain high values, even after the ten suspicious measurements were excluded from the weight-specific analysis.

Overall, these visualizations complement the descriptive statistics and demonstrate that extreme observations remain an important characteristic of the dataset.

Observations appearing beyond the boxplot whiskers are considered potential outliers according to the $1.5 \times$ IQR rule. However, these values are not necessarily data errors and were not automatically removed from the dataset.

5.3 Log Transformation of Review Counts

The number of customer reviews exhibited substantial positive skewness, with a small number of books receiving exceptionally high review counts.

To examine this distribution more clearly, a logarithmic transformation was applied:

$$Y = \log(1 + X)$$

where X represents the original number of reviews. Adding one allows books with zero reviews to be included, since $\log(0)$ is undefined. Figure 4 presents the distribution of log-transformed review counts.

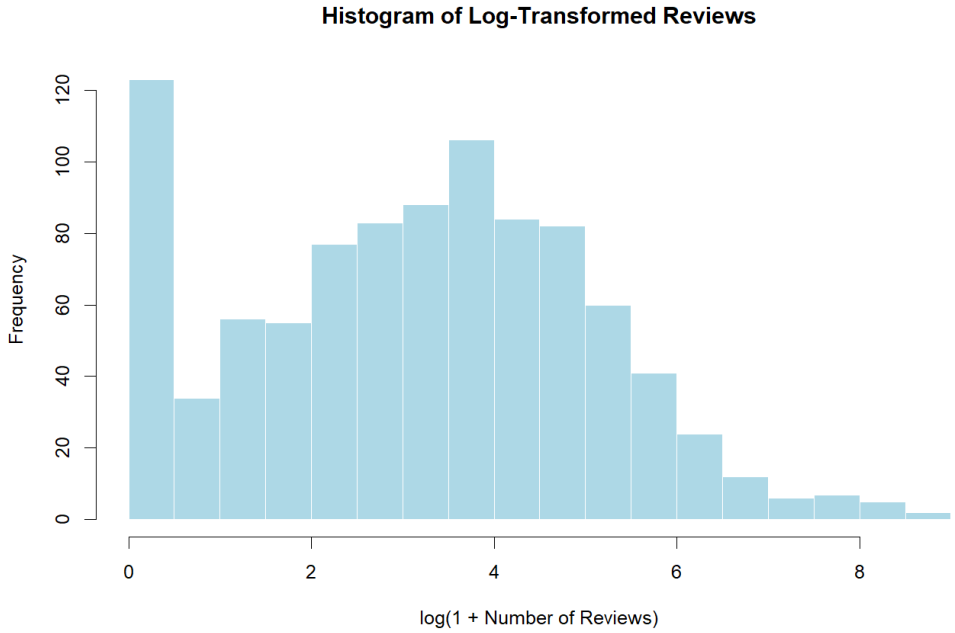


Figure 4: Histogram of Log-Transformed Review Counts.

The original review-count distribution had a skewness of 9.467, indicating a strong right tail. Following the transformation, skewness decreased to approximately 0.055, indicating substantially reduced asymmetry.

The transformation compresses large review counts while preserving their relative ordering. This makes the distribution easier to visualize and may be useful for subsequent statistical analysis. However, low skewness alone does not establish that the transformed data follow a normal distribution.

5.4 Log-Transformed Variables and Average Customer Ratings

To prepare the variables for the transformation, their distributions were first examined using the original weight measurements (`weight_kg`), including the ten potentially suspicious values (Figure 5). Compared with Figure 2, which uses the filtered weights, the weight distribution is much more strongly right-skewed because of a few extreme measurements.

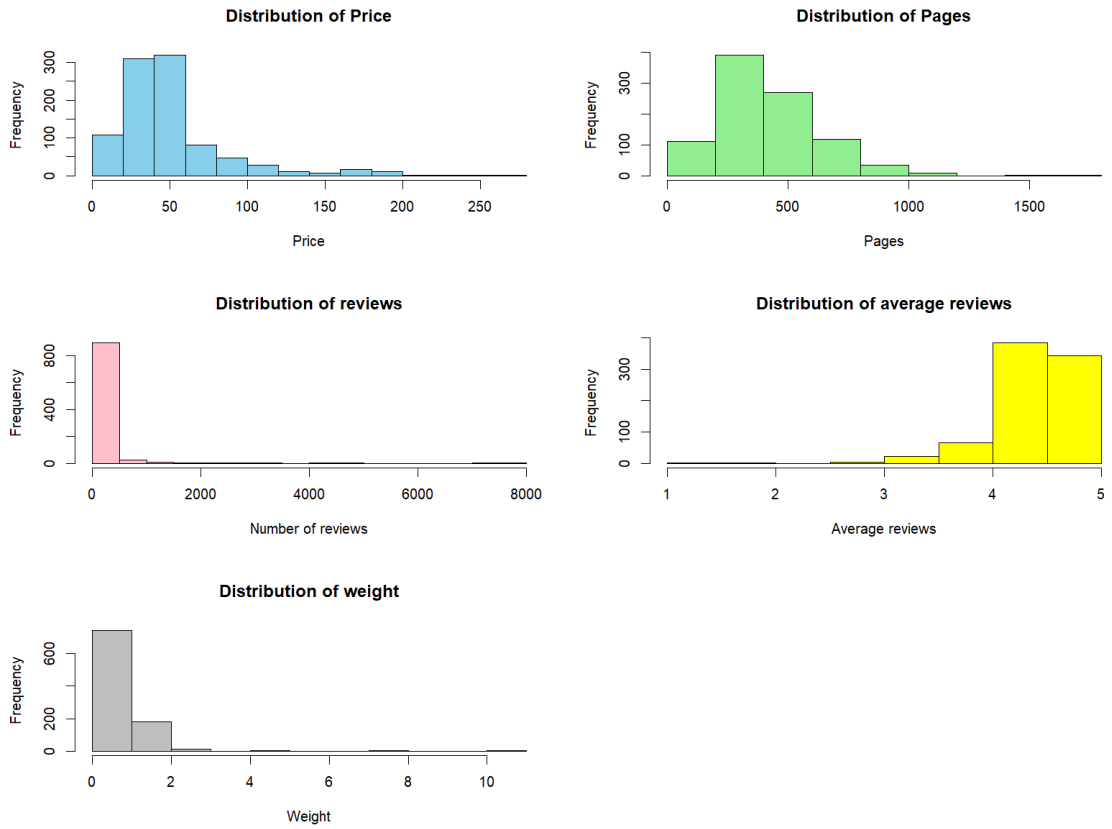


Figure 5: Distributions of Price, Page Count, Review Count, Average Customer Ratings, and Original Book Weight

Price, page count, review count and weight are right-skewed, while the average customer rating is the only left-skewed variable, indicating that customers tend to leave positive rather than negative reviews.

To reduce the skewness, the variables were transformed using $\log(1+x)$. Adding one is necessary because `n_reviews` contains zero values (123 books without reviews), for which $\log(0)$ is undefined. Such transformations can also be useful for subsequent hypothesis testing and model building.

To examine the effect of the transformation, the numerical variables were plotted against the average customer rating, first in their original form (Figure 6) and then after the log transformation (Figure 7). The average rating was chosen as a candidate response variable.

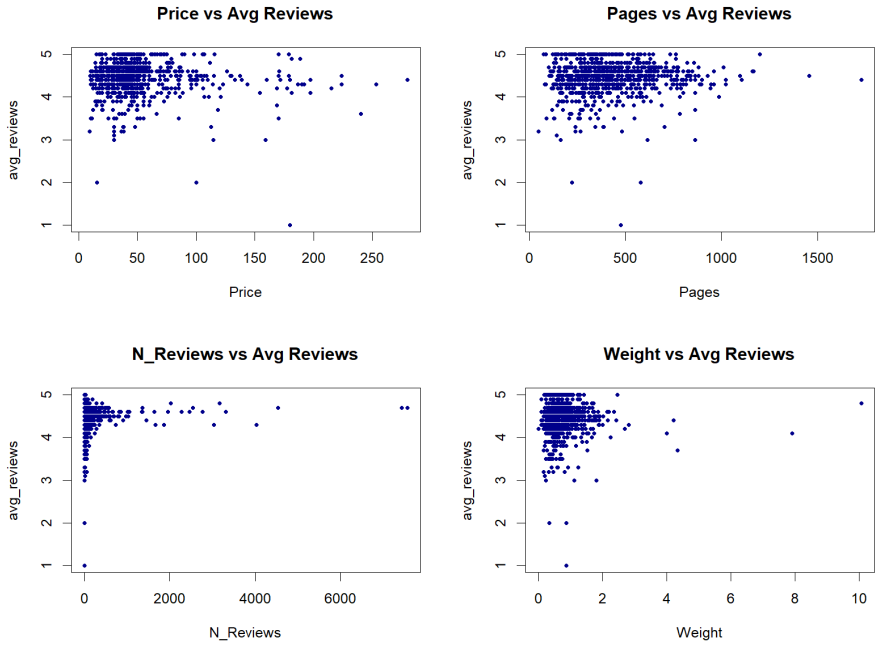


Figure 6: Original Numerical Variables versus Average Customer Ratings

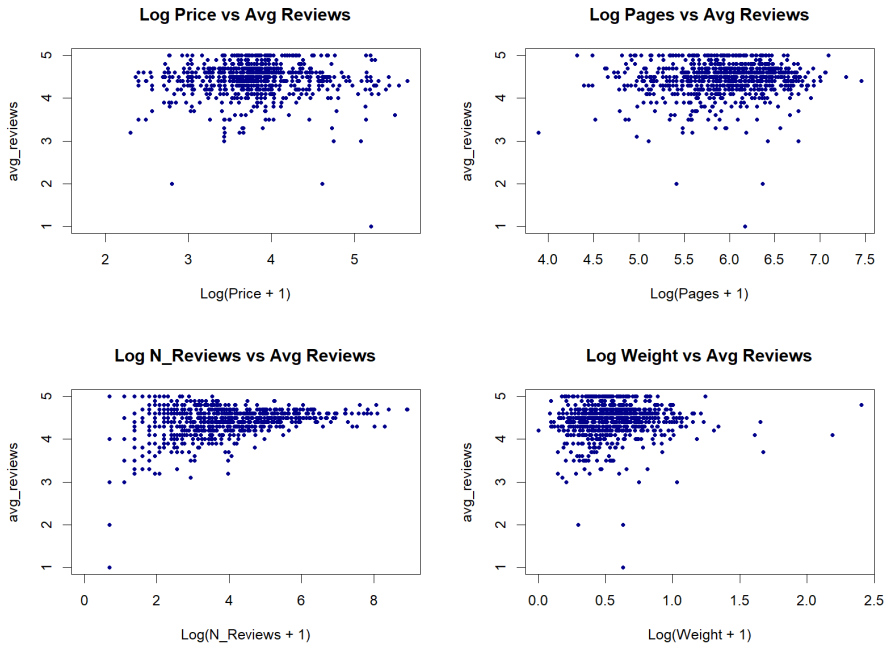


Figure 7: Log-Transformed Numerical Variables versus Average Customer Ratings

In the original scale, most observations are concentrated in the lower-left part of each plot because of a few extreme values. After the log transformation, the observations are spread more evenly, which makes the relationships with the average rating easier to examine. In both versions, no clear linear pattern between the average rating and price, pages, review count or weight is visible, which is consistent with the near-zero correlations reported in Section 5.8.

5.5 Star Rating Distribution

To further investigate customer evaluations, the average proportion of each category of star-ratings was calculated among the books with available customer reviews. Figure 8 presents the average distribution of ratings from one to five stars.

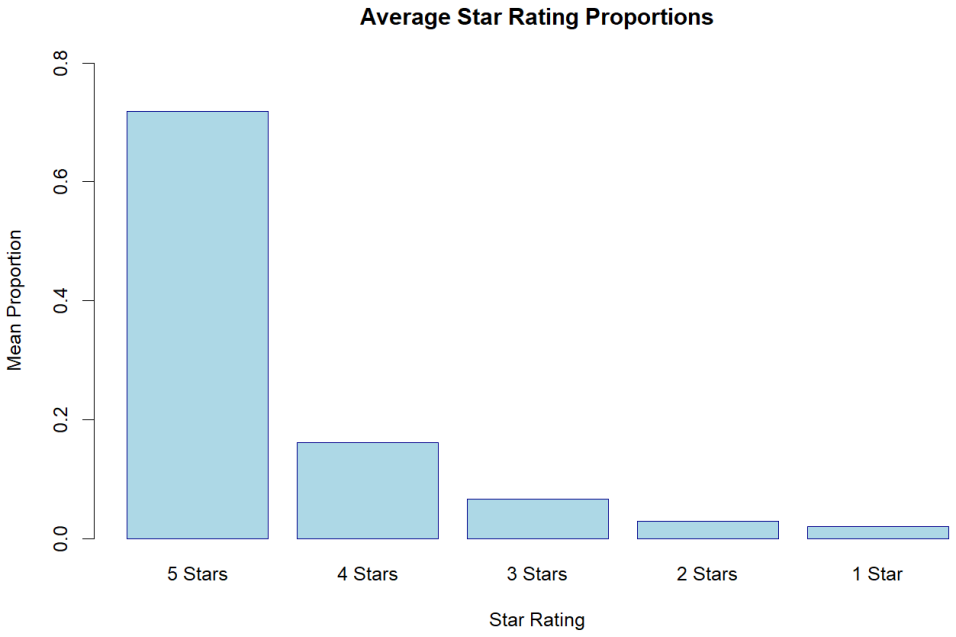


Figure 8: Average Distribution of Customer Star Ratings

Five-star ratings account for approximately 71.9% of the average rating distribution, followed by four-star ratings at 16.2%. Three-star ratings represent approximately 6.7%, while two-star and one-star ratings account for 3.1% and 2.2%, respectively.

These findings indicate that customer evaluations are predominantly positive, consistent with the high average ratings observed in descriptive statistics.

The percentages represent the mean rating proportions across books with customer reviews, with each book contributing equally, rather than the pooled distribution of individual customer reviews.

Variability in Star Rating Proportions

While Figure 8 summarizes the average proportion of each star-rating category, boxplots provide additional information about the variability of these proportions across books.

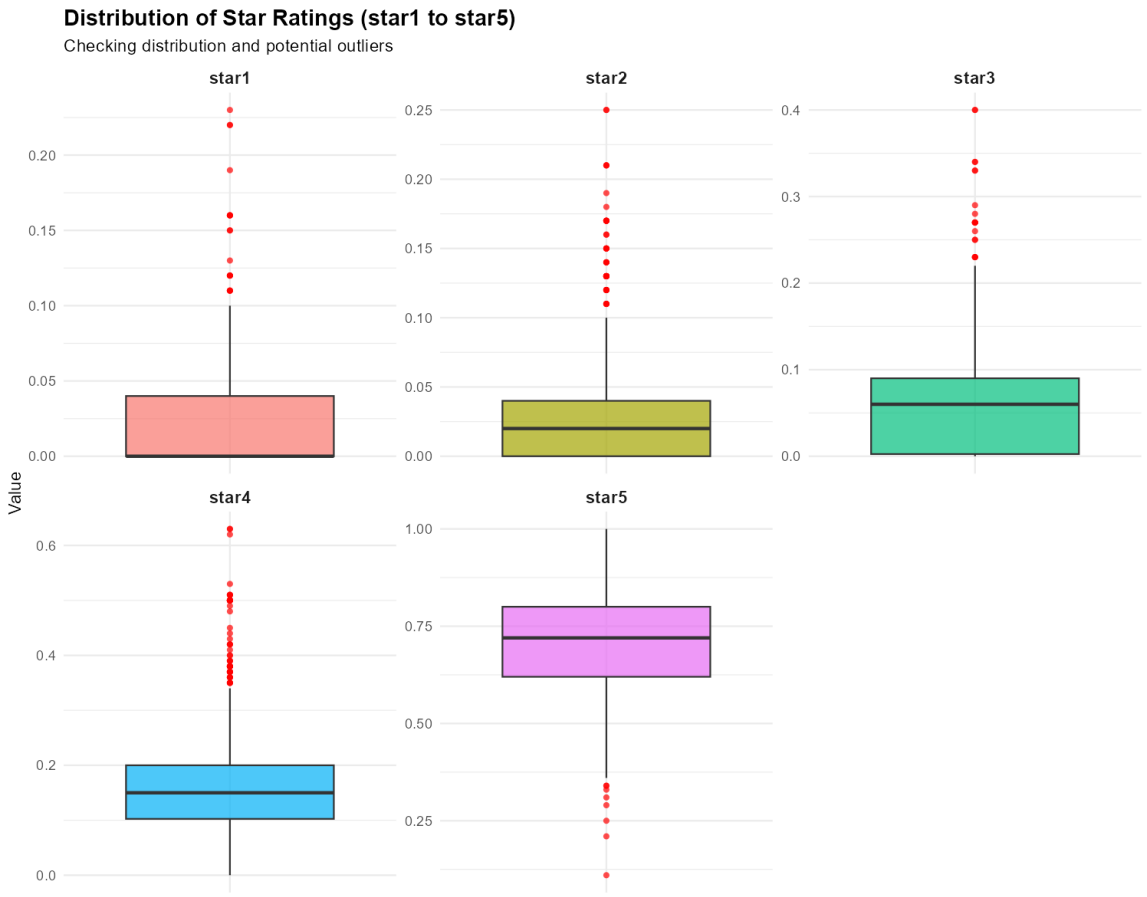


Figure 9: Distribution of Star Rating Proportions Across Books

The boxplots show that five-star ratings generally account for the largest proportion of reviews per book, with a median of approximately 0.72.

In contrast, one-star and two-star rating proportions are typically close to zero, while three-star and four-star proportions show greater variation.

Several observations appear beyond the boxplot whiskers. However, these values are not necessarily data errors, as unusually high or low rating proportions may reflect genuine customer evaluations.

Together, the bar chart and boxplots provide complementary information about the central tendency and variability of customer rating proportions.

Distribution of Star Rating Proportions

To further examine the distribution of customer evaluations, histograms with density curves were used to visualize the proportions of one-star through five-star ratings across books.

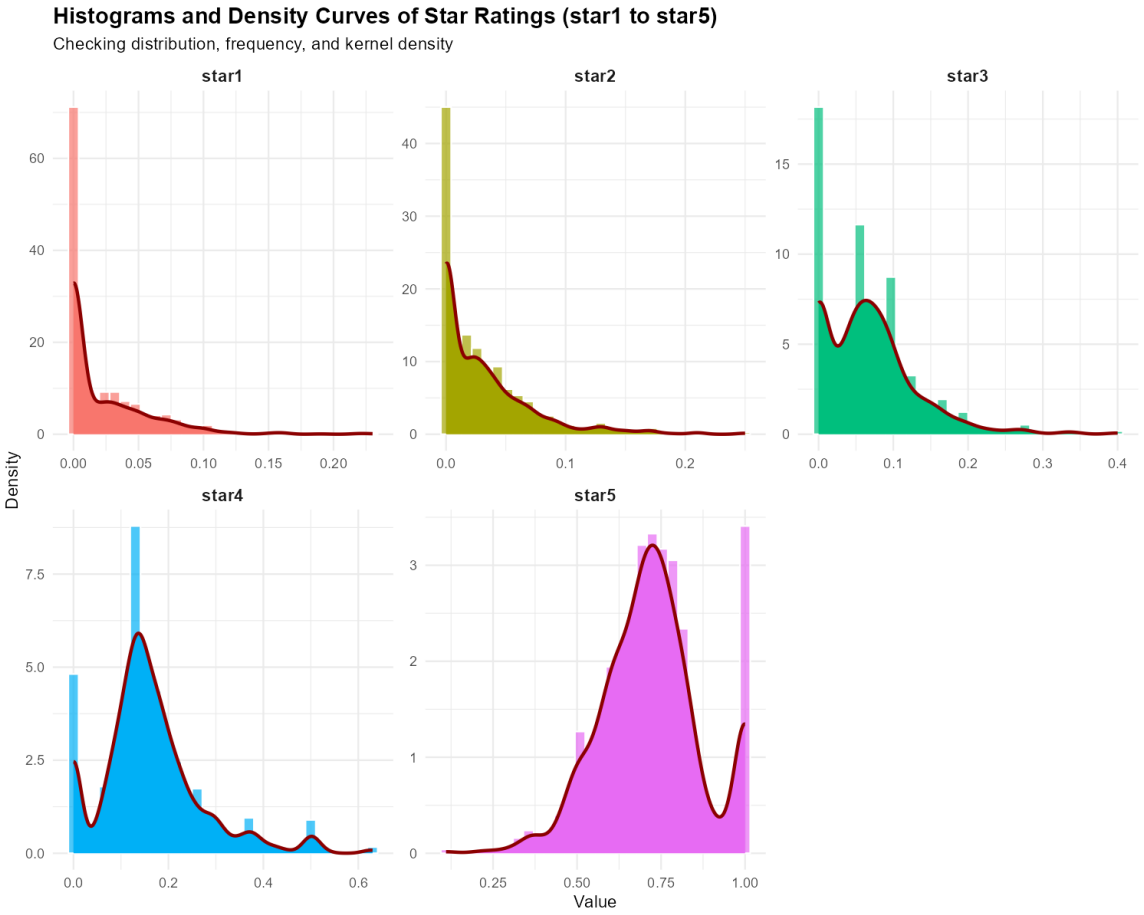


Figure 10: Histograms and Density Curves of Star Rating Proportions

Figure 10 shows that the proportion of five-star ratings is generally concentrated at relatively high values, with many books having proportions between approximately 0.60 and 0.85.

An additional concentration appears near 1.0, indicating that some books have almost exclusively five-star reviews.

In contrast, the distributions of one-star, two-star, and three-star rating proportions are strongly right-skewed, with many observations concentrated near zero.

Four-star rating proportions generally occupy an intermediate range, with considerable variation across books.

These patterns are consistent with the predominance of positive customer ratings observed in the previous visualizations.

Because the five star-rating proportions sum to approximately one for each reviewed book, their distributions should not be interpreted as independent.

5.6 Scatterplots and Relationships

Scatterplots were used to visually examine relationships between numerical variables. Figure 11 illustrates the relationship between the number of pages and book price.

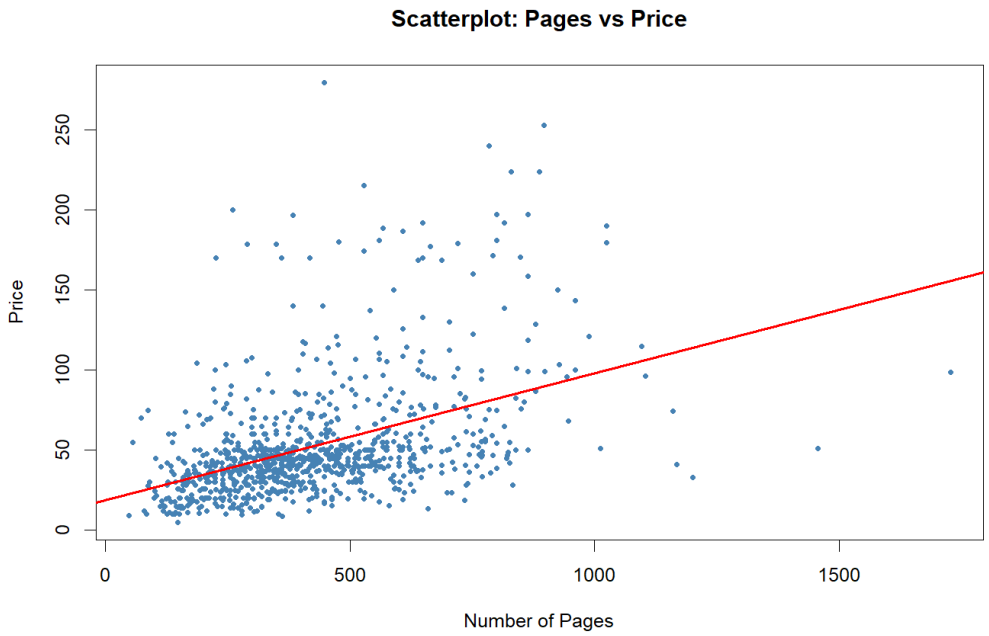


Figure 11: Relationship Between Page Count and Book Price

The scatterplot indicates a moderate positive relationship between page count and price, with a Pearson correlation coefficient of approximately 0.444 (931 books with both values available). Using the log-transformed price, $\log(1 + \text{price})$, the correlation increases to 0.486, which is the version tested formally in Hypothesis 1 (Section 6.1).

This suggests that books with more pages tend to be more expensive. However, considerable variation in price remains among books with similar page counts.

The observed association does not imply that increasing the number of pages directly causes a higher price.

The relationship between page count and book weight was examined separately in Section 3, where the effect of potentially suspicious weight measurements was also investigated.

5.7 Customer Reviews and Ratings

A scatterplot was also used to examine the relationship between average customer ratings and the number of reviews.

Since review counts are highly right-skewed, the log-transformed number of reviews was used to improve the visualization.

Figure 12 illustrates the relationship between average rating and log-transformed review counts.

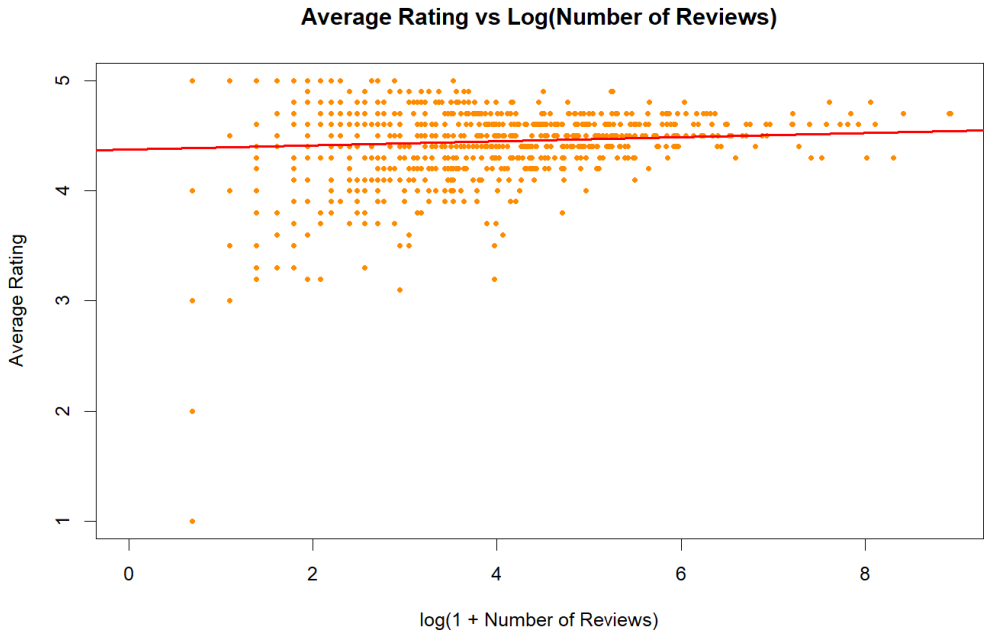


Figure 12: Relationship Between Average Rating and Log-Transformed Review Counts

The Pearson correlation coefficient between average rating and log-transformed review counts was approximately 0.074, indicating a very weak positive linear relationship.

The scatterplot suggests that books receiving more reviews do not necessarily have higher average ratings. Books with similar review counts may have substantially different ratings.

Overall, the number of reviews appears to have little linear association with the average customer rating in this dataset.

5.8 Correlation Matrix Heatmap

To analyse the correlations between the main numerical variables, a Pearson correlation matrix was constructed using price, pages, number of reviews, average customer ratings, and book weight.

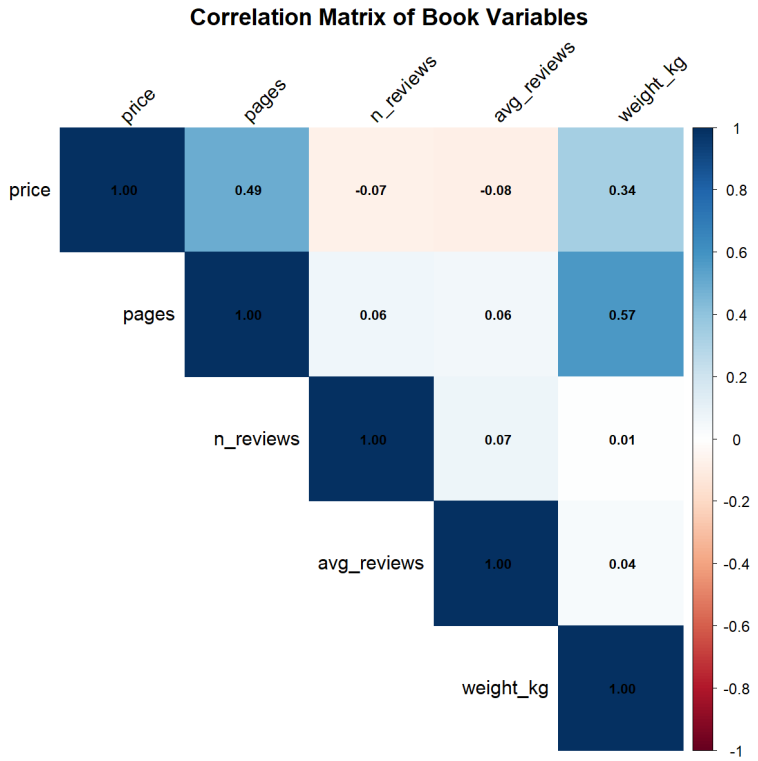


Figure 13: Correlation Matrix of Price, Pages, Review Counts, Average Ratings, and Weight

The matrix was computed using only the 811 books with complete information on all five variables, that is, books with at least one review and an available weight. For this subset, the correlation between price and page count is $r = 0.49$, slightly higher than the value of 0.444 obtained in Section 5.6 using all 931 books with available price and page count. In both cases the relationship is moderate and positive, which is consistent with longer books requiring more material.

A positive correlation is also observed between price and weight ($r = 0.34$), while pages and weight show a stronger positive correlation ($r = 0.57$).

However, average customer ratings show correlations close to zero with price, pages, number of reviews, and weight. These results suggest that the selected book characteristics have very weak linear associations with average customer ratings.

To extend this analysis, the following correlation matrix includes all numerical variables, including the individual star-rating proportions, both weight variables and the log-transformed variables.

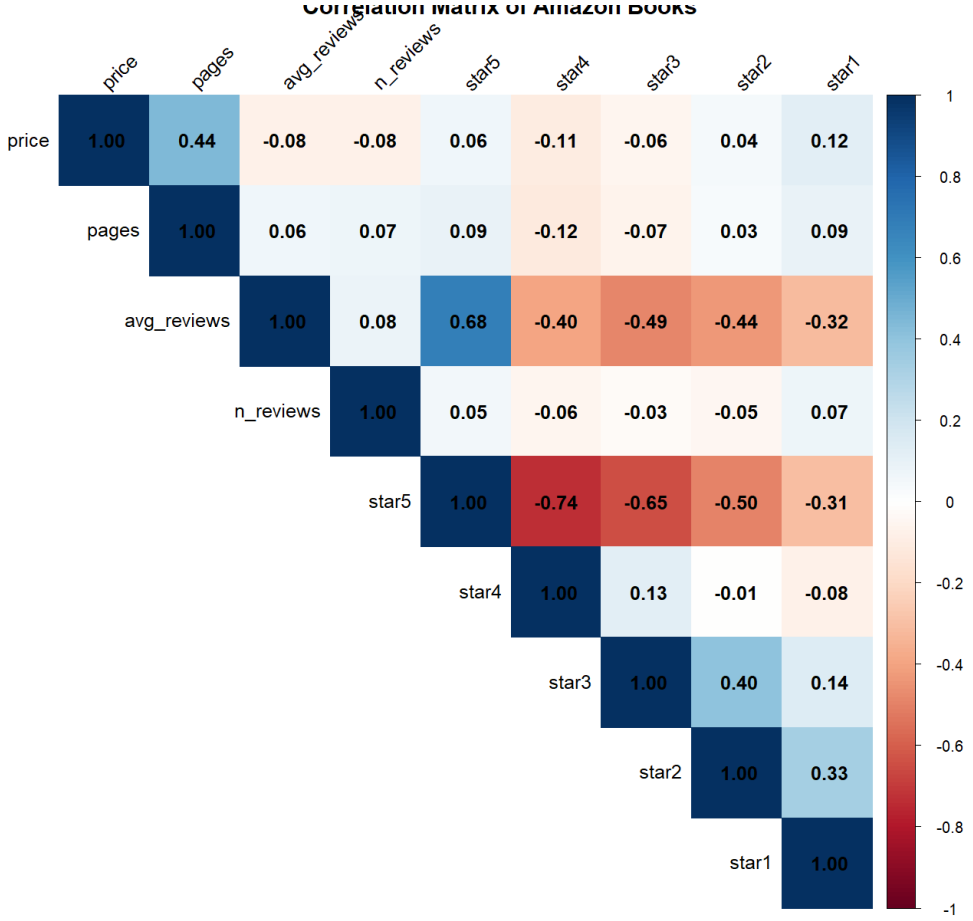


Figure 14: Pearson Correlation Matrix of Numerical Variables

Figure 14 uses pairwise complete observations, so each coefficient is based on all books with both values available. It shows a moderate positive correlation between book price and page count ($r \approx 0.44$), consistent with the corresponding scatterplot.

Average customer ratings show a strong positive correlation with the proportion of five-star reviews ($r \approx 0.68$). Conversely, the proportion of five-star reviews is negatively correlated with four-star ($r \approx -0.74$) and three-star ($r \approx -0.65$) review proportions.

These relationships should be interpreted carefully because the star-rating proportions sum to one and the average rating is mathematically related to their distribution.

In contrast, the number of customer reviews exhibits very weak linear correlations with both average ratings ($r \approx 0.08$) and book prices ($r \approx -0.08$).

Overall, the heatmap provides a broader view of the linear relationships within the dataset, complementing the more detailed scatterplot analysis.

5.9 Pairwise Scatter Plot Matrix

A pairwise scatter plot matrix was used to explore relationships among all numerical variables of the cleaned dataset: price, page count, average customer rating, review count, the five star-rating proportions, the original and filtered weights, and the log-transformed price, page count, review count and weight.

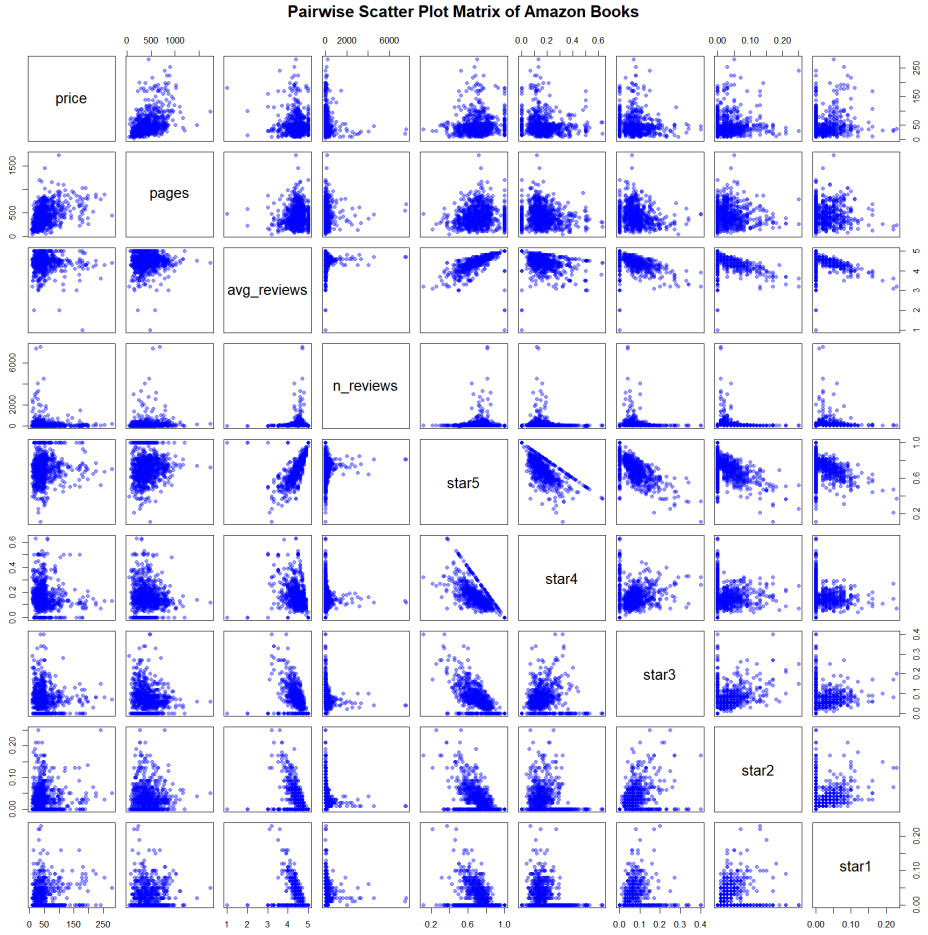


Figure 15: Pairwise Scatter Plot Matrix of Amazon Books

The scatter plot matrix provides a broader overview of the relationships between numerical variables. The plots involving review counts reveal a strong concentration of observations at relatively low values, together with a small number of books receiving exceptionally high numbers of reviews. The relationship between price and page count shows a moderate positive association, consistent with the individual scatterplot analysis. The star-rating proportions exhibit constrained patterns because the proportions for each reviewed book sum approximately to one.

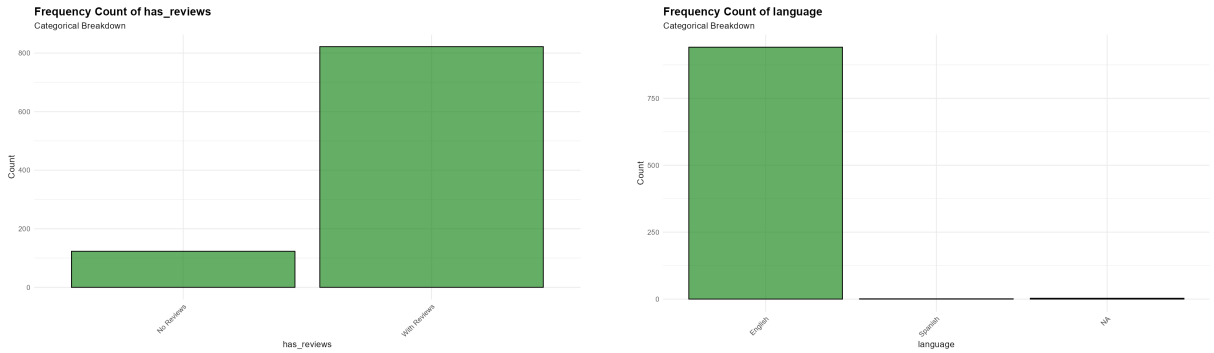
Overall, the pairwise scatter plot matrix complements the correlation heatmap by providing a visual overview of the relationships, clusters, and potential outliers in the dataset.

Because each log-transformed variable is a monotonic function of its original variable, the corresponding panels show near-perfect relationships by construction and should not be interpreted as substantive findings.

5.10 Categorical Analysis

Categorical variables were examined to investigate the distribution of customer review availability and book languages.

Two bar charts were used to visualize the frequency counts of the variables `has_reviews` and `language`.



(a) Availability of Customer Reviews (b) Distribution of Book Languages

Figure 16: Categorical Breakdown of Review Availability and Book Languages

Customer Review Availability

The dataset contains 945 books, of which 822 (approximately 87.0%) have at least one customer review, while 123 (approximately 13.0%) have no customer reviews.

This indicates that most books in the dataset have received customer feedback.

However, the absence of reviews for some books limits the availability of rating-related information for these observations.

Distribution of Book Languages

English is almost the only language in the dataset: 941 books (approximately 99.6%) are in English, one book is in Spanish, and the language is missing for 3 books.

Because the variable has almost no variation, it cannot support meaningful comparisons between language groups and was therefore excluded from the subsequent statistical analysis.

Overall, the categorical analysis complements the numerical exploration by highlighting differences in review availability and language representation across the dataset.

6 Hypothesis Testing and Empirical Findings

To formally evaluate the underlying relationships and market dynamics prior to econometric modeling, five primary statistical hypotheses were formulated and empirically tested.

6.1 Hypothesis 1: The Effect of Page Count on Book Price and Non-Parametric Modeling

- **Null Hypothesis (H_0):** There is no linear correlation between the number of pages (`pages`) and the log-transformed book price ($\rho = 0$).
- **Alternative Hypothesis (H_1):** Page count is linearly correlated with the log-transformed book price ($\rho \neq 0$).

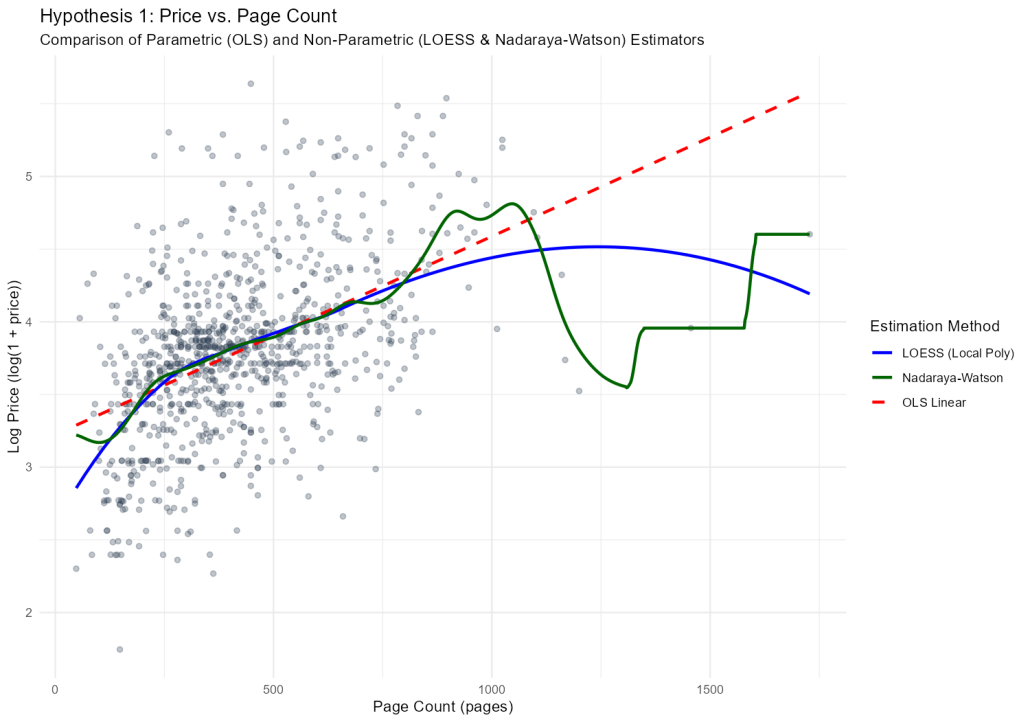


Figure 17: Hypothesis 1: Comparison of Parametric (OLS) and Non-Parametric (LOESS and Nadaraya–Watson) Estimators

Empirical Results and Statistical Testing

The Pearson product-moment correlation test revealed a statistically significant positive relationship between page count and $\log(1 + \text{price})$, with a correlation coefficient of $r = 0.4859$ ($t = 16.944$, $df = 929$, $p < 2.2 \times 10^{-16}$) and a 95% confidence interval from 0.4352 to 0.5335. Therefore, the null hypothesis is rejected.

Graph Interpretation and Non-Parametric Benchmarking

- In the region containing most observations (approximately 0–800 pages), the OLS line, the LOESS curve and the Nadaraya–Watson estimator closely agree, supporting a positive association between page count and log price.
- Above approximately 1,000 pages the data are sparse. Because the Nadaraya–Watson estimator is a local weighted average, it follows individual books in this region, producing sharp fluctuations and flat segments where no observations are nearby.
- The LOESS curve is smoother in this region, but its downward bend is also based on very few books. The OLS line extrapolates the linear trend into a region with almost no data.

Conclusion: Page count is positively associated with log price across the range where most books lie. Estimates for very long books (more than 1,000 pages) are unreliable because of data sparsity. As this is an observational association, it does not establish that additional pages cause higher prices.

6.2 Hypothesis 2: Price Differentiation Based on Review Engagement

- **Null Hypothesis (H_0):** The mean log-price of books without reviews (`No Reviews`) is equal to the mean log-price of books with active reviews (`With Reviews`).
- **Alternative Hypothesis (H_1):** There is a statistically significant difference in mean pricing between reviewed and unreviewed titles.

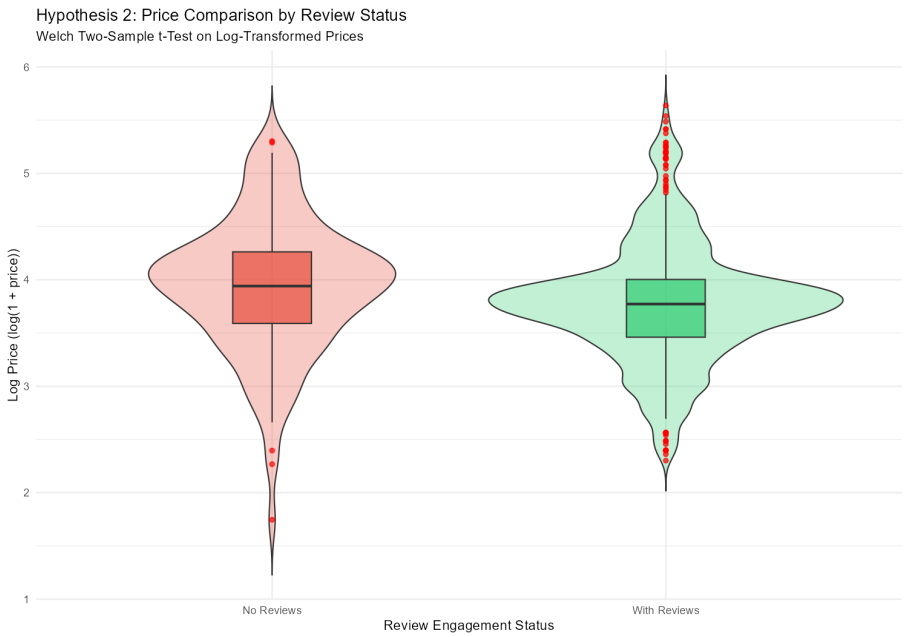


Figure 18: Hypothesis 2: Price Comparison by Review Engagement Status

Empirical Results and Group Estimates

A Welch two-sample t-test was conducted on $\log(1 + \text{price})$ to compare the two groups without assuming equal variances. The results show a statistically significant difference in means ($t = 2.7354$, $df = 151.67$, $p = 0.006973$). The 95% confidence interval for the difference ranges from 0.04620 to 0.28648, so the null hypothesis is rejected at $\alpha = 0.01$.

- **No Reviews Group:** Mean $\log(1 + \text{price}) = 3.943061$.
- **With Reviews Group:** Mean $\log(1 + \text{price}) = 3.776725$.

Interpretation: Books without reviews have a higher average log price than books with reviews. One possible explanation is that newly listed titles, which have not yet received reviews, are offered at their full list price, while popular titles are discounted over time. However, the dataset does not contain information on listing dates or discounts, so this explanation cannot be tested directly.

6.3 Hypothesis 3: Variance in Ratings Across Review Engagement Levels

Because the average rating variable (`avg_reviews`) is structurally missing (`NA`) for books with zero reviews, a direct comparison between unreviewed and reviewed books is unfeasible. Therefore, Hypothesis 3 evaluates rating dynamics within the subsample of reviewed books ($N = 822$), splitting them into High Engagement and Low Engagement categories based on the median review count.

- **Null Hypothesis (H_0):** The mean average review rating (`avg_reviews`) does not differ significantly between books with high and low review engagement.
- **Alternative Hypothesis (H_1):** Books with high review engagement exhibit significantly different average ratings compared to those with low engagement.

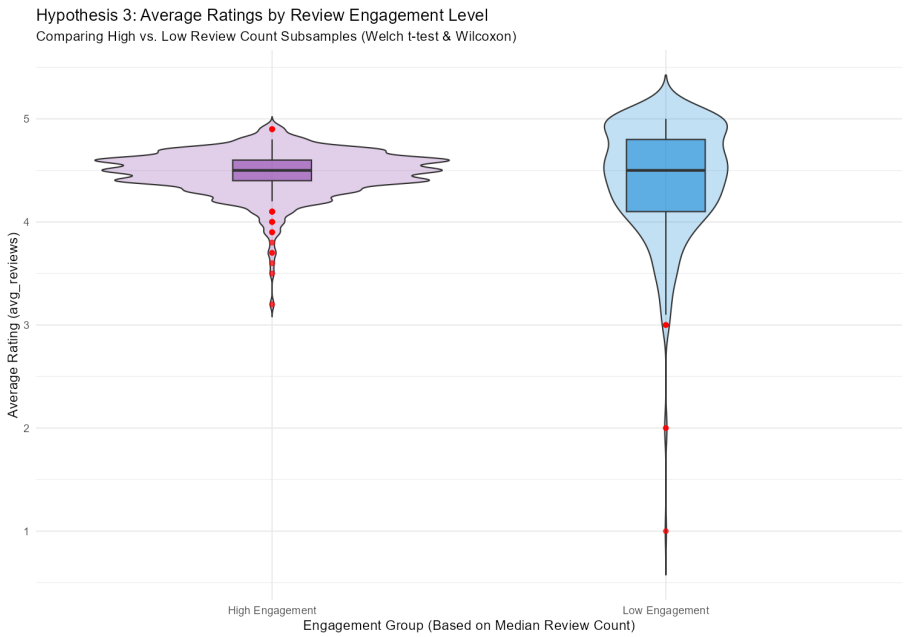


Figure 19: Hypothesis 3: Average Ratings Across Review Engagement Levels

Empirical Results and Methodological Conclusion

The parametric Welch two-sample t-test indicates a statistically significant difference in mean ratings between engagement groups ($t = 2.008, df = 534.01, p = 0.0452$), with the High Engagement group exhibiting a slightly higher mean rating ($\bar{x} = 4.469$) compared to the Low Engagement group ($\bar{x} = 4.413$).

However, the non-parametric Wilcoxon rank-sum test yields a non-significant result ($W = 83962, p = 0.8886$). Because the non-parametric test accounts for the severe negative skewness and extreme low-rating outliers in the distribution (particularly present in the Low Engagement subsample), we conclude that engagement volume does not exert a robust, distribution-free effect on book ratings once non-normality is controlled for.

These findings should also be interpreted in light of the rating consistency check, which identified potential discrepancies between reported average ratings and star-rating proportions. No observations were excluded solely on this basis.

6.4 Hypothesis 4: Review Volume vs. Book Pricing Dynamics

- **Null Hypothesis (H_0):** There is no monotonic correlation between the number of customer reviews (`n_reviews`) and the log-transformed book price.
- **Alternative Hypothesis (H_1):** There is a statistically significant monotonic correlation between review volume and book price.

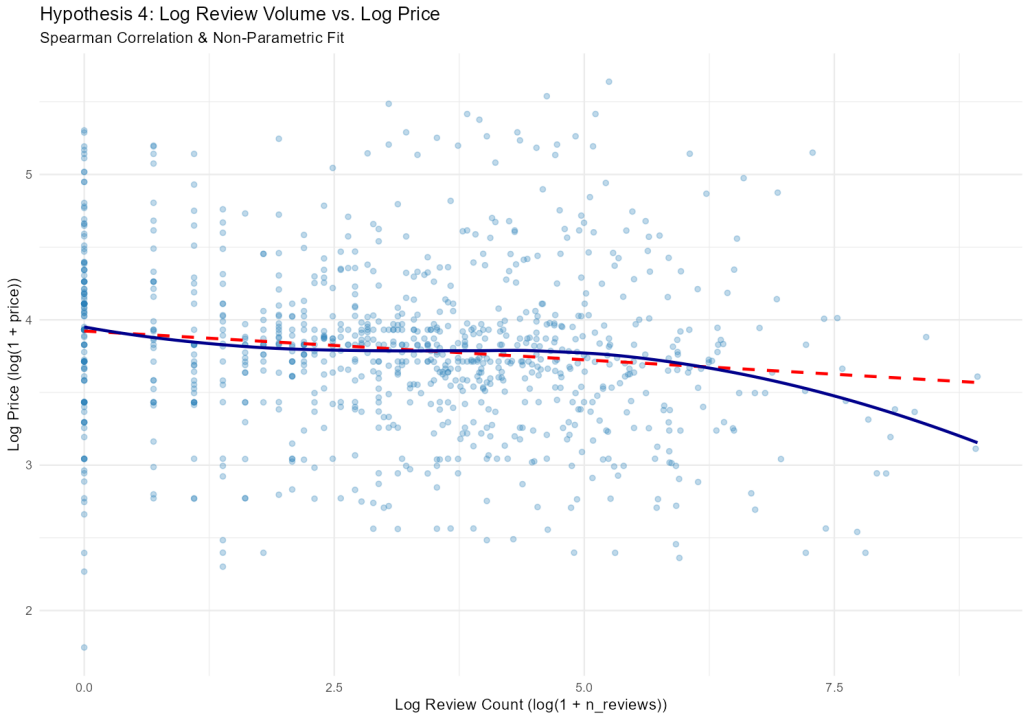


Figure 20: Hypothesis 4: Relationship Between Review Volume and Book Price

Empirical Results and Statistical Testing

The Spearman rank correlation test revealed a statistically significant negative association between the number of customer reviews and log-transformed book price ($\rho = -0.1561$, $S = 161572930$, $p = 1.465 \times 10^{-6}$).

The null hypothesis is rejected, indicating a weak but statistically significant negative monotonic relationship between review volume and book price.

6.5 Hypothesis 5: Average Customer Rating vs. Book Price

- **Null Hypothesis (H_0):** There is no monotonic correlation between the average customer rating (`avg_reviews`) and the log-transformed book price.
- **Alternative Hypothesis (H_1):** There is a statistically significant monotonic correlation between average customer ratings and book price.

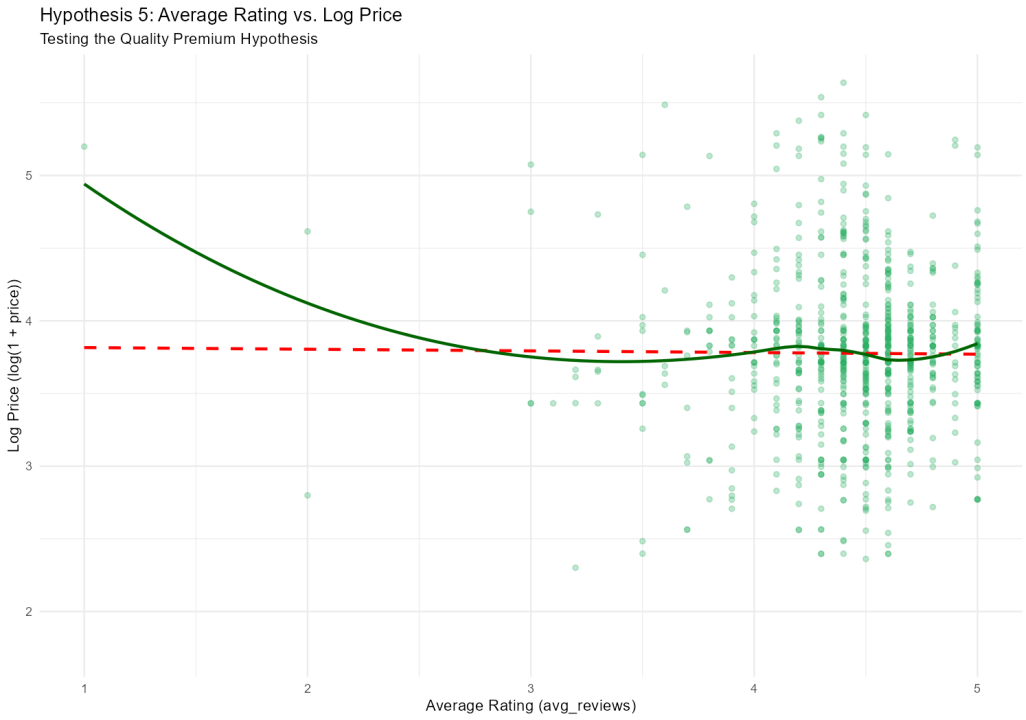


Figure 21: Hypothesis 5: Relationship Between Average Customer Rating and Book Price

Empirical Results and Statistical Testing

The Spearman rank correlation test revealed a negligible positive association between average customer ratings and log-transformed book price ($\rho = 0.01974$, $p = 0.5723$).

Since the p-value exceeds the conventional significance level of $\alpha = 0.05$, the null hypothesis cannot be rejected. The empirical findings suggest that average customer ratings do not exhibit a statistically significant monotonic association with book price.

This result should be interpreted cautiously because the rating consistency check identified potential discrepancies in 90 observations. These observations were retained in the analysis.

7 Multivariate Econometric Modeling and Diagnostic Evaluation

7.1 Multiple Linear Regression Model

To investigate the joint association between book characteristics and retail prices, a multivariate Ordinary Least Squares (OLS) regression model was estimated. The dependent variable was the log-transformed book price, while the independent variables included page count, review availability, and log-transformed review volume.

The econometric specification is expressed as:

$$\log(1 + \text{price}_i) = \beta_0 + \beta_1 \text{pages}_i + \beta_2 \text{has_reviews}_i + \beta_3 \log(1 + \text{n_reviews}_i) + \varepsilon_i \tag{1}$$

where β_0 represents the intercept, β_1, β_2, and β_3 represent the regression coefficients, and ε_i denotes the error term.

7.2 Empirical Regression Results

The estimated model was statistically significant, with an overall F-statistic of $F(3, 927) = 127.7$ and a p-value below 2.2×10^{-16}. The model was estimated on 931 books with complete information on all variables.

The model achieved an R^2 of approximately 0.2924 and an adjusted R^2 of 0.2901, indicating that approximately 29% of the variation in log-transformed book prices is explained by the included predictors.

Table 6: OLS Regression Results

Variable	Coefficient	t-value	p-value
Pages	0.001506	18.984	$< 2 \times 10^{-16}$
Has Reviews	0.004104	0.068	0.946
Log Review Count	-0.072200	-6.774	2.23×10^{-11}

7.3 Interpretation of Regression Coefficients

The page count coefficient is positive and highly statistically significant ($\beta_1 = 0.001506$, $p < 0.001$). Holding the other variables constant, each additional page is associated with approximately 0.15% higher price.

The review availability indicator is not statistically significant ($\beta_2 = 0.004104$, $p = 0.946$), suggesting that the presence of reviews does not explain price differences once page count and review volume are taken into account.

Log-transformed review volume has a statistically significant negative coefficient ($\beta_3 = -0.0722$, $p < 0.001$). Holding page count constant, a 1% increase in the number of reviews is associated with approximately 0.072% lower price.

The model explains about 29% of the variation in log price. The remaining variation is likely related to factors not available in the dataset, such as the publisher, binding type, author reputation and discounts.

7.4 Multicollinearity Assessment

Variance Inflation Factors (VIF) were calculated to evaluate potential multicollinearity among the independent variables.

Table 7: Variance Inflation Factors

Predictor	VIF
Pages	1.0461
Has Reviews	1.6475
Log Review Count	1.7016

All VIF values are substantially below the conventional threshold of 5, suggesting that severe multicollinearity is not present among the selected predictors.

7.5 Regression Diagnostic Evaluation

Regression diagnostic plots were examined to evaluate the behavior of model residuals, identify potentially influential observations, and assess the suitability of the OLS specification.

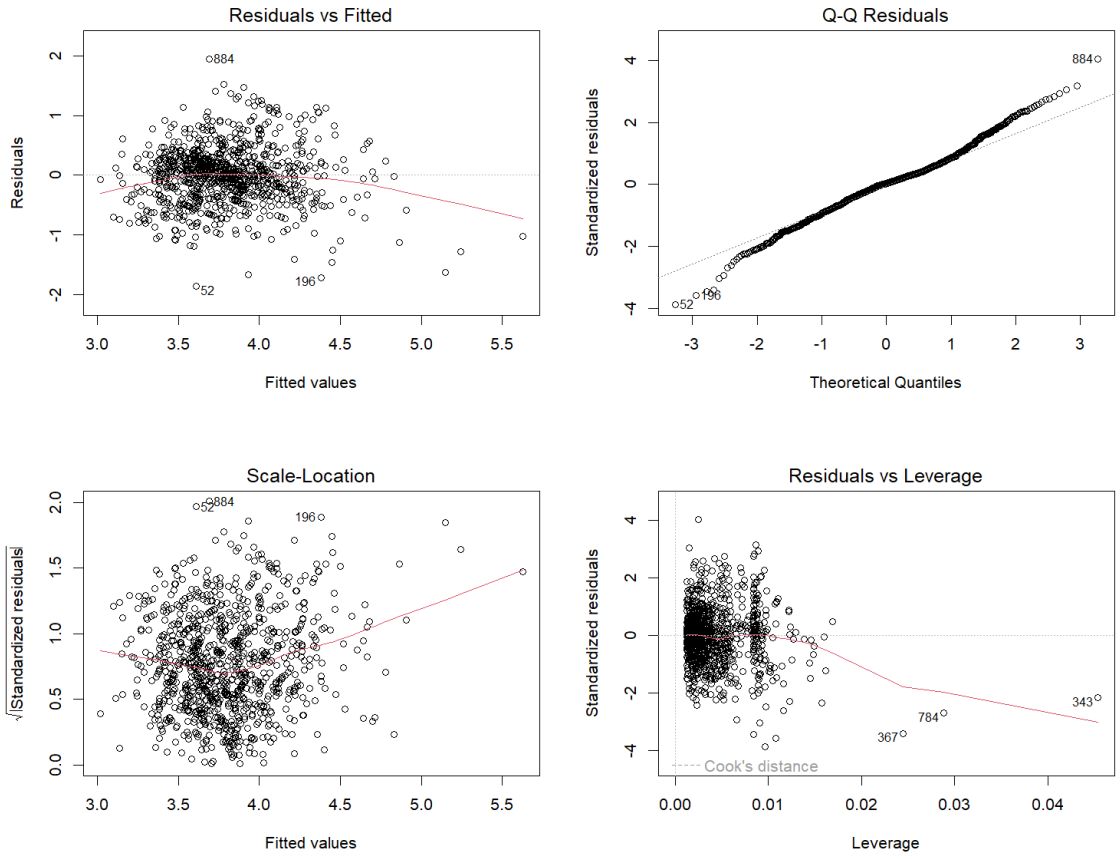


Figure 22: OLS Regression Diagnostic Plots

Assessment of OLS Diagnostic Plots

1. **Residuals vs Fitted (Linearity):** The residuals are generally centered around zero, although mild curvature suggests possible deviations from the linearity assumption.
2. **Normal Q-Q Plot (Normality):** The residuals approximately follow the theoretical quantiles in the central region, but deviations in the tails suggest departures from normality.
3. **Scale-Location Plot (Homoskedasticity):** The upward trend in residual spread at higher fitted values suggests possible heteroskedasticity. Therefore, constant error variance cannot be confidently assumed.
4. **Residuals vs Leverage (Influence):** Some observations exhibit relatively higher leverage or residual values and may warrant further investigation. The diagnostic plot alone does not establish the absence of influential observations.

Overall, the diagnostic plots suggest that some OLS assumptions may not be fully satisfied. Consequently, the regression results should be interpreted with appropriate caution.

8 Conclusion

8.1 Summary of Findings

This report analysed 946 Amazon data science books, of which 945 were retained after removing one duplicate. No books were removed because of extreme or missing values; missing values were handled using available-case analysis, and ten implausible weights were excluded only from the weight-specific analysis. The main findings are:

- **Page count is the strongest predictor of price.** It is positively correlated with log price ($r = 0.486$, $p < 2.2 \times 10^{-16}$) and remains significant in the regression, where each additional page is associated with approximately 0.15% higher price. The relationship is uncertain for books above about 1,000 pages, where data are sparse.
- **Books with more reviews tend to be cheaper.** Books without reviews have a higher mean log price ($p = 0.007$), and review volume is weakly negatively correlated with price ($\rho = -0.156$, $p < 0.001$). In the regression, the number of reviews remains significant, while the presence of reviews does not.
- **Ratings are not related to price.** No significant association was found ($\rho = 0.020$, $p = 0.572$), so there is no evidence of a quality premium.
- **Ratings do not differ robustly by review volume.** Welch's t-test suggested a small difference ($p = 0.045$), but the Wilcoxon test did not ($p = 0.889$).
- **Customer ratings are predominantly positive.** On average, about 72% of each book's ratings are five-star ratings.

8.2 Limitations and Possible Extensions

- The analysis is observational, so the associations cannot be interpreted as causal effects.
- The rating consistency check flagged 90 books, mainly with few reviews, whose reported rating differs from the one implied by the star proportions. They were retained, which may affect rating-related results.
- The regression explains about 29% of the variation in log price; factors such as publisher, binding type, author and discounts are not available.
- The diagnostic plots indicate non-normal residuals and heteroskedasticity, so the regression p-values should be interpreted with caution.
- Almost all books are in English, so the effect of language could not be studied.

Future work could use heteroskedasticity-robust standard errors, include variables such as publisher or publication date, and model non-linear effects of page count on price.

References

[1] Amazon Data Science Books dataset (`Amazon_Data_Science_Books.csv`), 946 observations and 11 variables. Provided by the course instructor for DSC531 – Statistical Simulation and Analysis, University of Cyprus.

[2] R Core Team (2026). *R: A Language and Environment for Statistical Computing*, version 4.6.1. R Foundation for Statistical Computing, Vienna, Austria. `https://www.R-project.org/`

[3] Wickham, H., Averick, M., Bryan, J., Chang, W., McGowan, L. D., François, R., et al. (2019). Welcome to the tidyverse. *Journal of Open Source Software*, 4(43), 1686.

[4] Wei, T. and Simko, V. (2024). *R package 'corrplot': Visualization of a Correlation Matrix*, version 0.95. `https://github.com/taiyun/corrplot`

[5] Fox, J. and Weisberg, S. (2019). *An R Companion to Applied Regression*, 3rd edition. Sage, Thousand Oaks, CA.

[6] Xie, Y., Allaire, J. J. and Grolemund, G. (2018). *R Markdown: The Definitive Guide*. Chapman and Hall/CRC.

[7] Tukey, J. W. (1977). *Exploratory Data Analysis*. Addison-Wesley, Reading, MA.

[8] Welch, B. L. (1947). The generalization of 'Student's' problem when several different population variances are involved. *Biometrika*, 34(1–2), 28–35.

[9] Cleveland, W. S. (1979). Robust locally weighted regression and smoothing scatterplots. *Journal of the American Statistical Association*, 74(368), 829–836.

[10] Nadaraya, E. A. (1964). On estimating regression. *Theory of Probability and Its Applications*, 9(1), 141–142.

[11] Watson, G. S. (1964). Smooth regression analysis. *Sankhyā: The Indian Journal of Statistics, Series A*, 26(4), 359–372.